

Multidimensional Goodhart

Response Channels and Residual Shape

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Abstract. Goodhart’s law is often summarized as proxy failure under optimization. This paper extract argues for a more structured object: under pressure, residual error has shape. Selection over a fixed baseline, intervention through fixed-type response, and multidimensional aggregation generate different distortion geometries. The useful transfer rule is therefore a response-modeling contract, not a universal theorem about dimensions or complexity.

1 Introduction: residual error has shape

The familiar slogan is that when a measure becomes a target, it ceases to be a good measure [4, 8]. The multidimensional version should not merely say that more proxy dimensions create more failure. The stronger and more brittle claim is that residual error has a response-dependent shape: direction, support, curvature, tail behavior, exchange rates, and dynamics.

Write the target-relevant state as $G(s) \in \mathbb{R}^m$, the proxy as $P(s) \in \mathbb{R}^k$, the intended proxy relation as $\varphi : \mathbb{R}^m \rightarrow \mathbb{R}^k$, and the residual as $\varepsilon(s) = P(s) - \varphi(G(s))$. A proxy can fail through hidden goal directions in $\ker \varphi$, through proxy residuals, or through agents moving after the score is used for control. The framework below separates those cases by the response channel.

The licensed thesis is not “more dimensions means more Goodhart” and not “optimization makes error generically more complex.” It is: a Goodhart claim should declare the response model that connects proxy pressure to target-relevant distortion.

2 Selection channels

In a pure selection channel, the policy reweights a fixed baseline law. Let $(S, \mathcal{F}, \mu)$ be the baseline space, $H : S \rightarrow \mathbb{R}^d$ hidden coordinates, and $P : S \rightarrow \mathbb{R}$ the scalar proxy. A selection policy has nonnegative weight W_θ with finite positive expectation, likelihood ratio $L_\theta = \frac{W_\theta}{\mathbb{E}_\mu[W_\theta]}$, selected law $\mu_\theta = L_\theta \mu$, and hidden response

$$B_{H(\theta)} = \mathbb{E}_{\mu_\theta}[H] - \mathbb{E}_{\mu[H]}.$$

Hard thresholds use $W_t = \mathbb{1}\{P \geq t\}$ and define the threshold response $b_{H(t)} = \mathbb{E}[H \mid P \geq t] - \mathbb{E}[H]$. Boltzmann selection uses $W_\beta = \exp(\beta P)$ only on the finite-mgf domain $\mathcal{B} = \{\beta : \mathbb{E}_{\mu[\exp(\beta P)]} < \infty\}$. Where differentiation is valid,

$$d\mathbb{E}_{\beta[H]} / d\beta = \text{Cov}_{\beta(H, P)},$$

so covariance is a local velocity along a selection path, not a finite-pressure primitive. The toy example $P = Z$, $H = Z^2 - 1$ has zero baseline covariance but nonzero threshold and finite Boltzmann response.

The Gaussian threshold model gives the cleanest coordinate picture. If $X \sim \mathcal{N}(0, \Sigma)$, $P = X_1$, and $H = (X_2, \dots, X_m)$, then

$$\mathbb{E}[H \mid X_1 \geq t] = \sigma_1 \lambda(t/\sigma_1) r,$$

where $r_j = \frac{\text{Cov}(X_j, X_1)}{\text{Var}(X_1)}$. Dimensional growth appears only if the coupling-vector norm grows with the number of hidden coordinates; a bounded total coupling budget blocks it.

Selection bounds. If $\delta = \|L_\theta - 1\|_{L^2(\mu)} = \sqrt{\chi^2(\mu_\theta \parallel \mu)}$ and hidden coordinate standard deviations are s_i , then

$$|B_{H_i}(\theta)| \leq \delta s_i, \quad \|B_{H(\theta)}\|_2 \leq \delta \|s\|_2.$$

For a declared value vector v ,

$$|\Delta(v \cdot H)| \leq \delta \sqrt{v^T \Sigma_H v}.$$

The first form is coordinate-explicit Euclidean drift. The second is declared value/operator drift; it is the right form when the scalar welfare direction is part of the model.

These are baseline-only claims. They stop at reweighting. They do not analyze what fixed agents can do after a metric is announced.

3 Intervention channels

Intervention begins when the policy changes behavior at fixed type. Declare a type space U , baseline type law ν , baseline kernel $K_0(ds | u)$, participation weights $W_{\theta(u)}$, and policy-indexed response kernels $K_{\theta(ds|u)}$. The induced law is

$$\mu_{\theta(A)} = \left(\int W_{\theta(u)} K_{\theta(A|u)} \nu(du) \right) / \left(\int W_{\theta(u)} \nu(du) \right).$$

Pure selection relative to (U, K_0) has $K_\theta = K_0$ almost everywhere and all policy dependence in W_θ . Intervention has $K_\theta \neq K_0$ on a positive-mass set of types. Mutual singularity with baseline is sufficient evidence that baseline-only selection fails, but it is not the definition: absolute continuity can survive when the baseline already contains epsilon mass on the response image.

The smallest algebraic intervention toy is a Stackelberg threshold game. An agent has true quality Q , can choose action $a \geq 0$, receives score $Q + a$, passes if $Q + a \geq t$, gains prize V , and pays cost $\frac{a^2}{2\kappa}$. In the noiseless case, agents with $Q \in [t - \Delta, t)$ game just enough to pass, with

$$\Delta = \sqrt{2\kappa V}.$$

The wedge is controlled by ease of gaming and stakes, not by baseline covariance. It is a toy signature of quadratic threshold gaming, not a law of neural training or RLHF.

For a more general local intervention budget, declare an action a , convex cost $c(a)$, linear score gain $w \cdot a$, and score deficit d . The score-deficit cost is

$$m(d) = \sup_{\lambda \geq 0} [\lambda d - c^*(\lambda w)],$$

under the usual finite-dimensional convex regularity. Gaming the deficit is privately feasible when $m(d) \leq V$. This is not a welfare bound until a hidden harm functional has also been declared.

4 Multidimensional scorecards

Adding measured dimensions has no sign by itself. In an additive scorecard with measured set M , separable quadratic costs $\sum_{j \in M} \frac{a_j^2}{2\kappa_j}$, score $\sum_{j \in M} w_j a_j$, and hidden harm $\sum_{j \in M} h_j a_j$, the cost-minimal action for fixed deficit d is

$$a_j^* = d \kappa_j w_j / W_M, \quad W_M = \sum_{i \in M} \kappa_i w_i^2,$$

and fixed-deficit per-agent harm is

$$H_{\text{per}(M,d)} = d \cdot \left(\sum_{j \in M} h_j \kappa_j w_j \right) / \left(\sum_{j \in M} \kappa_j w_j^2 \right).$$

Thus $H_{\text{per}(M,d)}$ is conserved across active measured sets if and only if $h_j = cw_j$ on the channels being compared. In the unit-weight equal-harm case, re-routing conserves fixed-deficit per-gamer harm, but increasing the aggregate gaming capacity can expand entry and raise population harm $H_{\text{pop}(M,F_Q,V)}$.

A conjunctive scorecard flips the per-gamer comparison. If passing requires $a_j \geq t$ for every $j \in M$, then $H_{\text{per}}^{\text{conj}(M,t)} = t|M|$ under equal harm. The population sign still depends on entry, because the cost of passing also rises. The contrast is the point: aggregation and exchange rates determine the distortion geometry.

5 Response-modeling contract and MMLU

The surviving framework is methodological. A claim should declare: type space, baseline behavior, policy exposure, response channel, action/search geometry, proxy and target, aggregation rule, hidden harm, and evidence standard. This contract is the transfer rule for applications. Domains do not inherit toy bounds by analogy; they inherit the obligation to declare the primitives that make a calculation or falsifier meaningful.

MMLU is a useful worked example because one score can support several different channels [6]. Selecting among fixed checkpoints is a selection claim over candidates and may use W_θ plus declared hidden value weights. Finetuning, prompt search, contamination, synthetic data filtering, and reward/proxy optimization are response-kernel claims and require K_θ plus action/search geometry. Model-selection overfitting [1], adaptive holdout reuse [2], strategic classification [5], performative prediction [7], and reward-model overoptimization [3] are nearby formal warnings, but none by itself supplies κ , V , a convex cost, or a welfare functional.

MMLU is Layer-2 application discipline, not empirical proof. It shows how the contract separates fixed-checkpoint selection from finetuning/search/contamination/reward-proxy mechanisms. It does not license Stackelberg, convex-cost, RLHF, or welfare bounds without additional primitives and falsifiers.

6 Open problems

Several problems remain deliberately open.

- **Welfare-bound packaging.** Convex private affordability and realized cost-minimizing harm are distinct from worst-case welfare bounds for a declared hidden-harm functional.
- **Endogenous value and stakes.** V may depend on observed gaming, trust, institutional response, or future proxy repair.
- **Adaptive hardening.** Hardening the currently exploited channel may converge, cycle, or expand the attack surface; it needs dynamic models.
- **Empirical implementation.** Applications need repeated-type observations, randomized or staggered exposure, action traces, cost variation, or pre-registered hidden outcomes.
- **Non-convex ML and RLHF mappings.** Gradient accessibility, benchmark contamination, reward-model simplicity, and optimizer search are not interchangeable substitutes for κ or convex costs.

Iterations 20–21.1 provide deterministic Layer-3 checks for the toy regimes: linear-Gaussian and nonlinear selection response, finite-mgf Boltzmann behavior, declared value vectors, the Stackelberg wedge, quadratic water-filling, noisy response diagnostics, welfare-bound separation, active-face quadratic repairs, capped fixed-charge switches, and additive/conjunctive population

checks. They are toy checks of the framework’s bookkeeping, not empirical confirmation of a general Goodhart law.

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